PD 125

# Prevalence and Characteristics of Asymmetric Hearing Loss in Korean Adults: A KNHANES 2022–2023 Analysis

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## Introduction

- **Asymmetric Hearing Loss (AHL)** is defined by interaural threshold difference.
- National epidemiologic data on AHL in Korean adults remain limited.
- This study evaluated prevalence and audiometric characteristics of AHL using **KNHANES 2022–2023**.
- This study aims to estimate the **prevalence of AHL** in Korean adults aged ≥40 years and to **characterize** its age, frequency, laterality, and degree patterns.

## Materials & Methods

- **Definition**: Absolute interaural difference ≥15 dB HL in 4-frequency average (0.5, 1, 2, 4 kHz), based on the AAO–HNS criterion
- Adults aged **≥40 years**
- Final sample: N = 7258
  - Weighted overall population: ~ 25.2 million
- AHL case subset: n = 606

## Results ▶ Key Findings

**8.0%** — Weighted prevalence of AHL

**4 kHz** — Most common asymmetry frequency

**L > R** — Left-sided AHL more common than right

**15–20 dB HL** — Largest category of interaural difference

### Weighted Prevalence of AHL by Age Group (N = 7258)

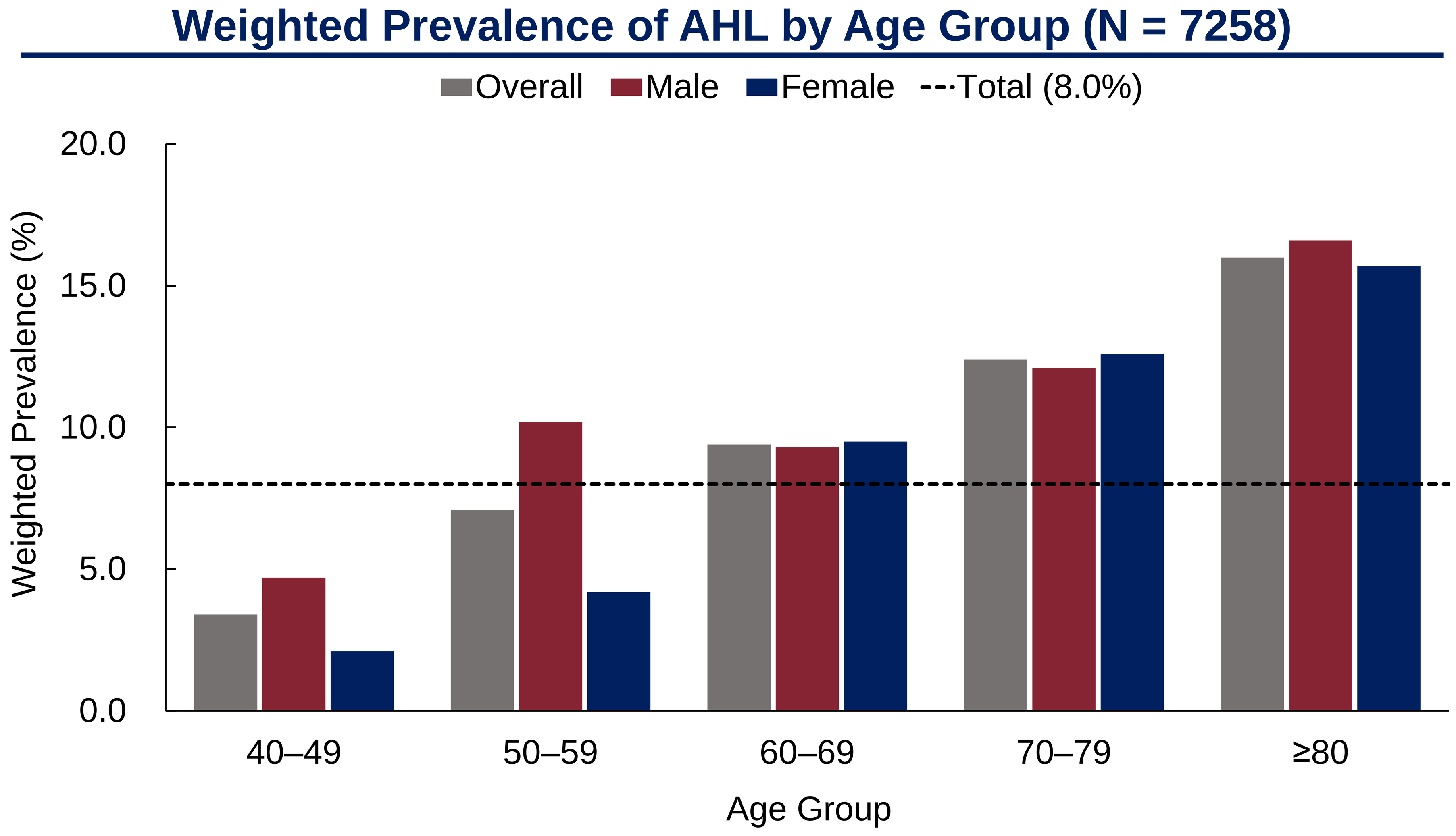


**Figure 1.** Weighted prevalence of AHL increased with age overall and in both sexes. Dashed line indicates the total prevalence of 8.0%

### Weighted Mean Hearing Thresholds by Frequency (N = 7258)

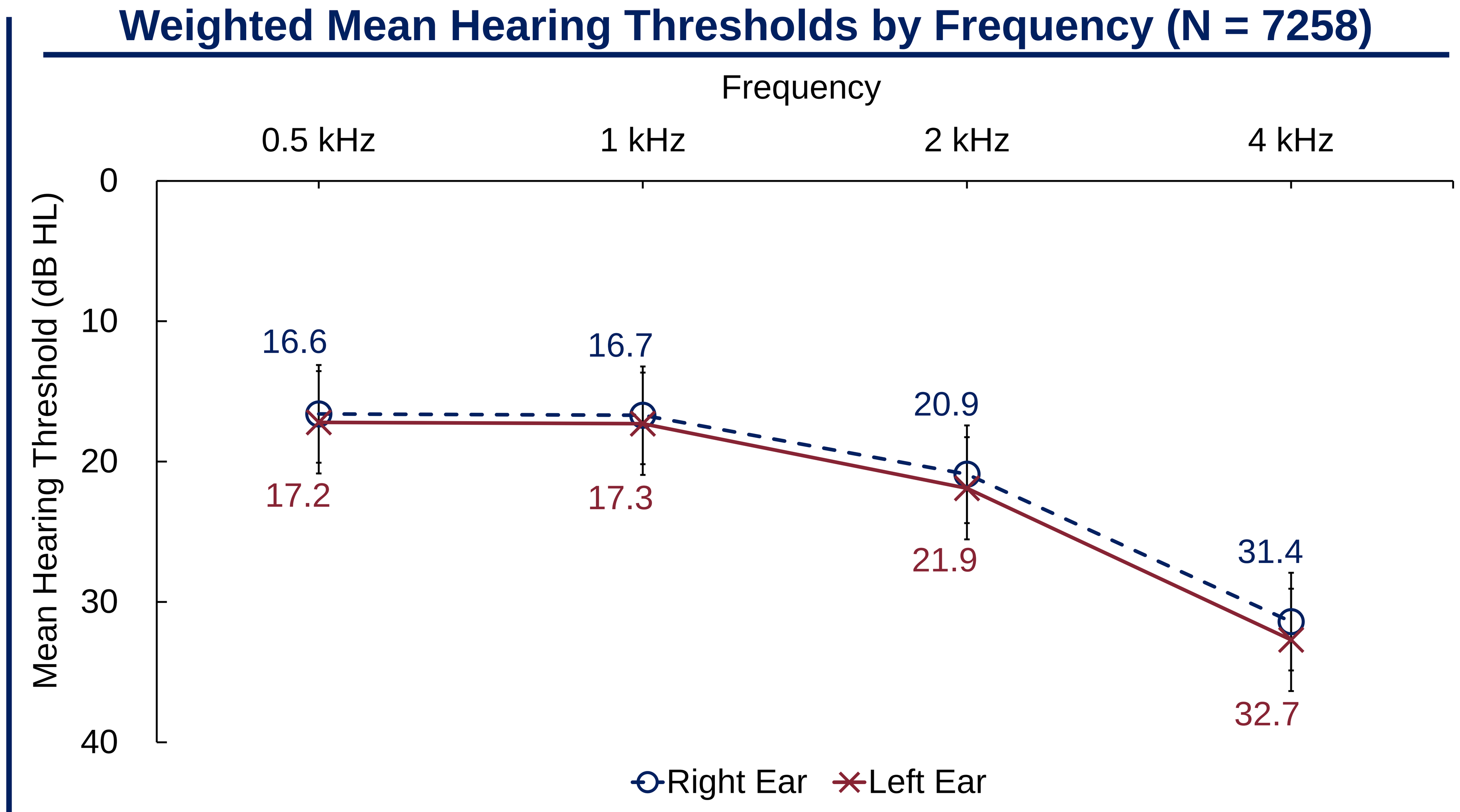


**Figure 2**. Left-ear thresholds were higher than right-ear thresholds across all frequencies, with the interaural difference more pronounced at 2 and 4 kHz.

### Frequency-Specific Asymmetry (n = 606)

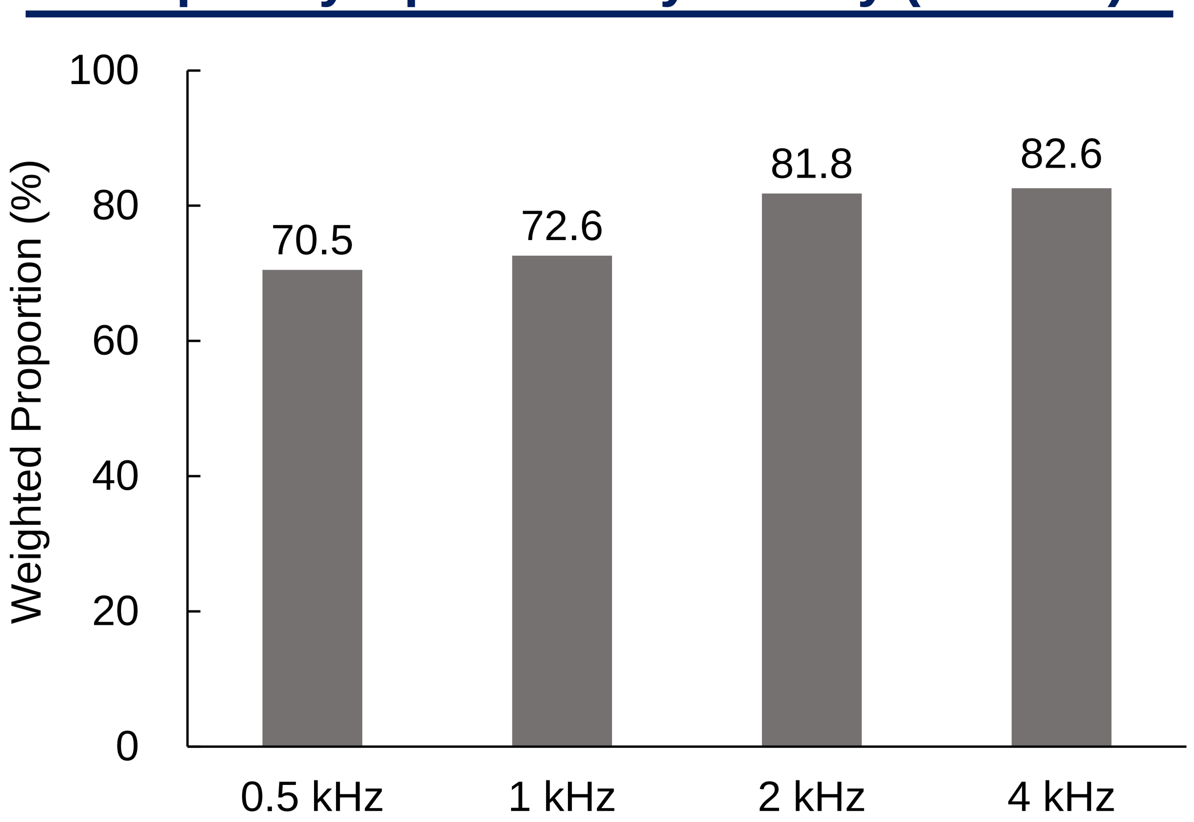


**Figure 3.** Asymmetry increased with frequency. Proportions are not mutually exclusive and do not sum to 100%, because participants could have asymmetry at more than one frequency.

### Laterality of AHL (n = 606)

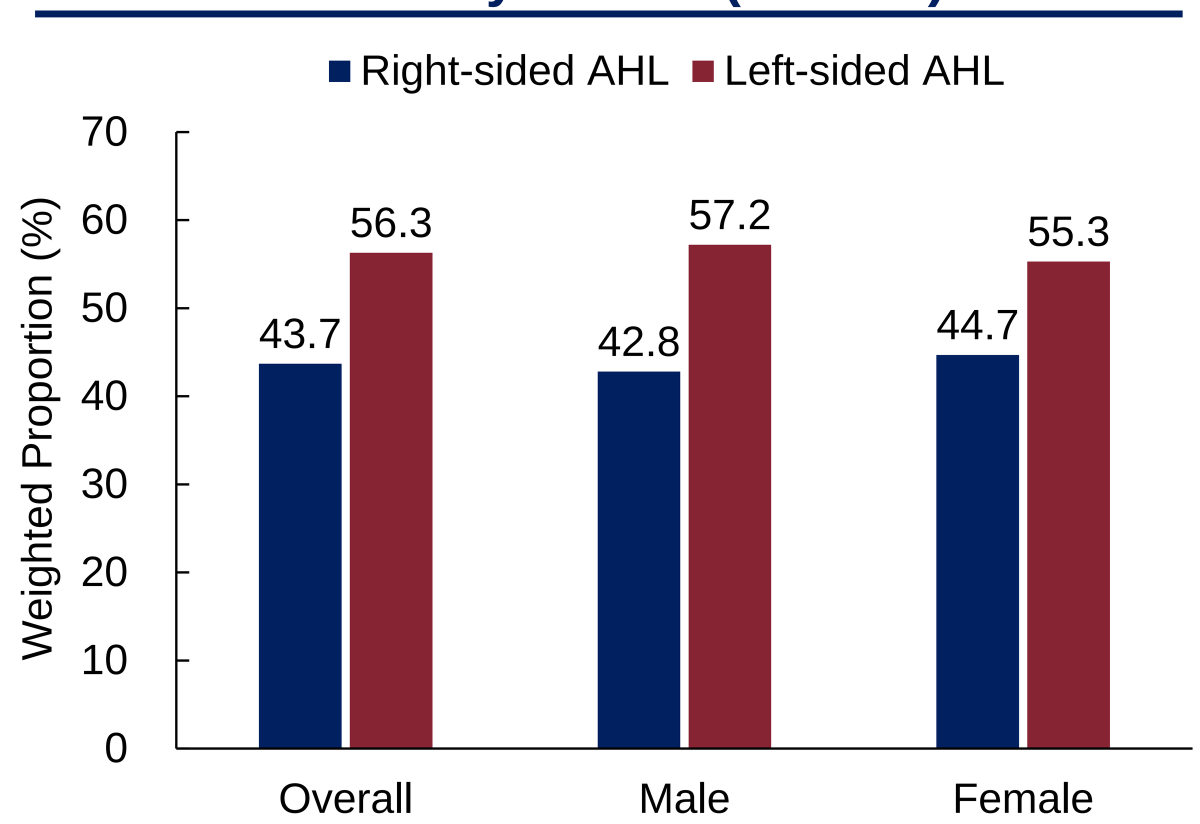


**Figure 4.** Left-sided AHL was more common than right-sided AHL overall and in both sexes, indicating a consistent left-sided predominance across sex subgroups.

### Interaural Differences by Age (n = 606)

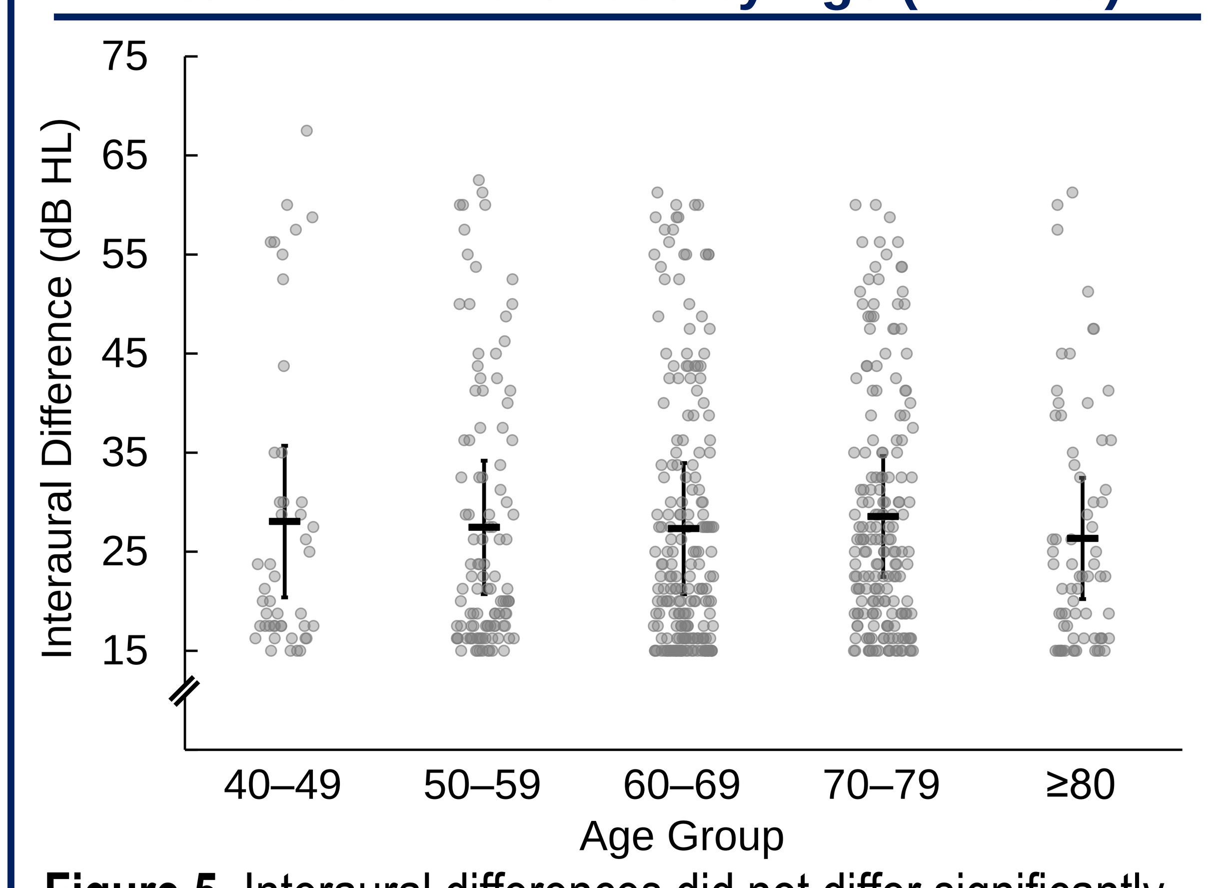


**Figure 5.** Interaural differences did not differ significantly across age groups, with most cases clustered within the 15–20 dB HL range. Horizontal bars and error bars indicate mean ± ½ SD.

## Conclusion

**In Korean adults aged ≥40 years**, AHL affected **8.0%** of the population and generally increased with age overall and in both sexes. **Older age** and **male sex** were independently associated with AHL. AHL was characterized by **higher-frequency asymmetry**, **left-sided predominance**, and clustering near the lower end of the asymmetry range, with **most cases falling within the 15–20 dB HL range.** These findings suggest that AHL is most often expressed as relatively mild interaural asymmetry rather than marked asymmetry.

## Resources

**SCAN FOR POSTER & REFERENCES**

Access the full poster, high-resolution figures, and reference links via QR code

WCA 2026 | International Society of Audiology ISA | The Korean Audiological Society



The authors declare no conflicts of interest.